Centralizers of a primitive action on pairs

Human Compression project: a component of the L044 recovery

Let $\mathcal{B}$ be a partition of $2m$ points into m pairs, where $m > 2$. Its full stabilizer is $V \rtimes S_m$, with $V = \mathbb{F}_2^m$ acting by independent flips. We first prove a statement about the actual permutation action, rather than assuming that a commuting permutation induces a permutation of $\mathcal{B}$.

Lemma 1. *Suppose $B \leq V \rtimes S_m$ is transitive on $\mathcal{B}$ and its induced group P is primitive and generated by involutions. Then either every element of $C_{S_{2m}}(B)$ preserves $\mathcal{B}$, or B is regular on the $2m$ points, m is an odd prime, and $P \cong B$ is dihedral of order $2m$. In the latter case an independent generating family of involutions for P has at most two members.*

Proof. Put $N = B \cap V$. If $N \neq 1$, some coordinate projection of N is nontrivial. Conjugation by B permutes its coordinate projections transitively, so every one is nontrivial. Its point orbits are consequently exactly the pairs of $\mathcal{B}$. A permutation centralizing B centralizes each element of N and therefore permutes these N-orbits. This proves the claim in that case; no characteristic-subgroup assertion is needed.

Suppose $N = 1$, and identify B with P via the block action. A point orbit of B meets every pair, and the size of that intersection is constant. There are either two orbits, each meeting every pair once, or one orbit of size $2m$. In the first case each row action is the literal faithful primitive action P on $\mathcal{B}$. The centralizer of that action is trivial. Indeed the orbits of a nontrivial centralizer would form a P-invariant block system; primitivity would make that centralizer transitive, forcing P to be regular. A regular primitive group is cyclic of prime order, and for $m > 2$ is not generated by involutions. Thus an equivariant permutation of either row is the identity. An equivariant bijection between the two rows is the unique bijection matching their points in each original pair: compose any such bijection with that one to obtain a centralizer of one row. Every permutation commuting with B therefore preserves the pairs.

In the transitive case fix a point x in a pair Δ. Write $J = B_x$ and $M = B_\Delta$. The group M is maximal in B, because the block action is primitive, and $[M : J] = 2$. Hence $J \trianglelefteq M$ and $M \leq N_B(J)$. Maximality gives $N_B(J) = M$ or $N_B(J) = B$. For a faithful transitive action, the centralizing permutations are the equivariant maps of the coset action on B/J; their possible images of J are precisely the cosets in $N_B(J)/J$. If this normalizer is M, every such map sends x within Δ, and equivariance sends every pair to itself.

If the normalizer is B, then $J \trianglelefteq B$. Faithfulness of the point action forces $J = 1$, so B is regular and $|M| = 2$. Let a generate M. Some involution b of B lies outside M, since the involutions generate B. Maximality of M forces $B = \langle a, b \rangle$, a dihedral group. Its rotation subgroup has order m. A proper nontrivial divisor of m would give a dihedral subgroup strictly between M and B, contrary to maximality. Thus m is prime, and $m > 2$ makes it odd. Two distinct involutions generate this entire dihedral group, so a family of at least three involutions cannot be independent. □

Theorem 2. *Let $G = A_{2m} = \langle \rho_0, \ldots, \rho_{r-1} \rangle$ in its natural action, with a string C-marking, and let $m \geq 8$. Suppose $H = G_i$ is transitive and preserves a maximal block system consisting of m pairs. If its primitive block image P is different from A_m, S_m, then $r \leq m - 1$. Moreover, in every putative case $r \geq m$, regardless of P, the connected block-generating interval L used below has at most one remaining noncommuting neighbour in H.*

Proof. Choose an inclusion-minimal distinguished family L whose images generate P. Its images are independent and nonidentity involutions. Write C for the marks of H outside L commuting with all of L, and R for the other marks. We acquire the full L047 maximal-block decomposition. If the noncommuting diagram of the images of L is disconnected, then $m \geq 60$, $|L| \leq 2\log_2 m$, $|C| \leq 1$ and $|R| \leq 4$. It follows that

$$r = 1 + |L| + |C| + |R| \leq 2\log_2 m + 6 < m.$$

The last inequality holds at $m = 60$ and its right-hand difference is increasing for $m \geq 60$.

Otherwise L is an interval of the full string, $|C| \leq 1$, and $|R| \leq 2$. Put $B = \langle L \rangle$ and assume $r \geq m$. The exceptional alternative of Lemma 1 would give $|L| \leq 2$ and hence $r \leq 6 < m$. Thus every permutation centralizing B preserves the chosen pairs. The omitted ρ_i cannot centralize B: together with H it generates A_{2m}, which preserves no nontrivial block system. It therefore fails to commute with some mark of L. By the string relations, i is adjacent to that interval. One of the only two possible noncommuting neighbours of L is consequently absent from H, so $|R| \leq 1$ and

$$r \leq |L| + 3. \tag{1}$$

This proves the final assertion for every primitive block image.

For $P \neq A_m, S_m$, the accepted correction to the source's Proposition 3.3 applies to the projected family: an independent generating string of involutions in a primitive group of degree $m \geq 8$, other than A_m, S_m, has at most $m - 4$ members. Its statement does not require the intersection property in the quotient. Thus $|L| \leq m-4$, and (1) gives $r \leq m-1$, a contradiction to the supposed exception. This proves the theorem. □

Inputs and scope. The source is Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Section 3.2. The L047 block decomposition and its precise classical inputs remain acquired. The projected string bound is the accepted corrected theorem, with its full finite proof and declared software premises; the source's false bound for arbitrary independent families is not used. Lemma 1 supplies the missing justification for attaching the omitted label to L in this pair-block branch. It uses the actual block kernel and the normalizer of a point stabilizer; it does not treat a block-breaking permutation as already acting on the blocks. The cases $P = A_m, S_m$ remain separate, and the degree-twelve and fourteen boundaries are covered by acquired complete finite classifications. No new computation, whole-original credit, conjecture answer or Lean verification is asserted here.